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Here is a list of common SCCM (System Center Configuration Manager) ports along with their protocols:

SCCM Ports and Protocols

Client Communication

- HTTP: TCP 80
- HTTPS: TCP 443

Site Server Communication

- SQL Server: TCP 1433 (default), TCP 4022 (replication)
- SQL Server Browser Service: UDP 1434
- RPC Endpoint Mapper: TCP 135
- RPC Dynamic Ports: TCP 49152-65535

Management Point

- HTTP: TCP 80
- HTTPS: TCP 443

Distribution Point

- HTTP: TCP 80
- HTTPS: TCP 443
- SMB: TCP 445

Software Update Point

- HTTP: TCP 80
- HTTPS: TCP 443
- WSUS: TCP 8530 (HTTP), TCP 8531 (HTTPS)

Reporting Services Point

- HTTP: TCP 80

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- **HTTPS: TCP 443**
- **SQL Server Reporting Services: TCP 80, TCP 443**

Client Push Installation

- **File and Printer Sharing: TCP 445**
- **Windows Management Instrumentation (WMI): TCP 135**

Asset Intelligence Synchronization Point

- **HTTP: TCP 443**
- **SQL Server: TCP 1433**

Endpoint Protection

- **HTTP: TCP 80**
- **HTTPS: TCP 443**

SCCM, or **System Center Configuration Manager**, is a tool developed by Microsoft to manage and deploy software, hardware, and other resources across an organization's network. It is part of the Microsoft System Center suite of products and is now known as **Microsoft Endpoint Configuration Manager**.

Key Features of SCCM:

- **Software Deployment:** Allows IT professionals to remotely install, update, and manage software across multiple devices.
- **Hardware Management:** Helps in managing hardware inventory and configurations.
- **Operating System Deployment:** Facilitates the deployment of operating systems on a large scale.
- **Security and Compliance:** Ensures that devices comply with security policies and standards.
- **Monitoring and Reporting:** Provides tools for monitoring the health and performance of the network and generating detailed reports.

SCCM integrates with various Microsoft technologies, such as **Microsoft Intune**, **Azure**, and **Windows Server Update Services (WSUS)**, to provide a comprehensive management solution.

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In SCCM (System Center Configuration Manager), there are three main types of sites, each serving different purposes and scales of deployment:

1. Central Administration Site (CAS)

- **Purpose:** Acts as the top-level site in a hierarchy, primarily used for large-scale deployments.
- **Uses:** Provides a central point of administration and reporting for the entire hierarchy. It does not directly manage clients but oversees multiple primary sites.
- **Scenarios:** Ideal for organizations with a large number of devices spread across various geographical locations.

2. Primary Site

- **Purpose:** Manages clients directly and can be either a stand-alone site or part of a hierarchy under a CAS.
- **Uses:** Handles client management tasks such as software deployment, updates, and compliance settings.
- **Scenarios:** Suitable for medium to large organizations. A stand-alone primary site can be used for smaller deployments but can also be expanded into a hierarchy with a CAS if needed.

3. Secondary Site

- **Purpose:** Extends the capabilities of a primary site, particularly in managing network bandwidth and client communications in remote locations.
- **Uses:** Acts as a local point for content distribution and client management, reducing the load on the primary site.
- **Scenarios:** Useful for managing clients in remote or branch offices where network bandwidth is a concern

Each type of site plays a crucial role in ensuring efficient and scalable management of devices and resources within an organization.

The **Assets and Compliance** tab in SCCM (System Center Configuration Manager) is a crucial workspace for managing and monitoring various aspects of your IT environment. Here's a breakdown of its main components:

Assets

- **Devices:** Lists all the devices managed by SCCM, including details like device name, status, and last activity. You can perform actions such as remote control, software deployment, and more.

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- **Users:** Displays user information and allows you to manage user-based deployments and settings.
- **User Collections:** Groups of users that you can manage collectively for software deployment, compliance settings, and more.
- **Device Collections:** Groups of devices that you can manage collectively, similar to user collections.

Compliance

- **Compliance Settings:** Allows you to create and manage configuration items and baselines to ensure devices comply with your organization's policies.
 - **Configuration Items:** Define specific settings and rules that devices must comply with, such as security settings, software configurations, and more.
 - **Configuration Baselines:** Collections of configuration items that you can deploy to devices to assess and enforce compliance.
- **Company Resource Access:** Manages access to company resources like VPN profiles, Wi-Fi profiles, and certificates.
- **Endpoint Protection:** Integrates with Microsoft Defender to manage and monitor endpoint security settings and compliance.

Monitoring and Reporting

- **Compliance Reports:** Provides built-in reports to monitor the compliance status of devices and configuration baselines.

The **Assets and Compliance** tab is essential for maintaining the health, security, and compliance of your IT infrastructure.

The **Software Library** tab in SCCM (System Center Configuration Manager) is a central hub for managing software-related tasks. Here's an overview of its main components:

Application Management

- **Applications:** Create, manage, and deploy applications to devices and users. This includes setting up deployment types, dependencies, and requirements.
- **Packages:** Manage traditional packages and programs, which are useful for deploying scripts, administrative tools, or legacy applications.
- **Software Updates:** Manage and deploy software updates to ensure devices are up to date with the latest patches and security updates.
- **Operating System Deployment:** Create and manage task sequences for deploying operating systems to new or existing devices.

Software Metering

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- **Metered Software Usage:** Track and report on the usage of software applications across your organization to optimize licensing and ensure compliance.

Compliance Settings

- **Configuration Items and Baselines:** Create and manage configuration items and baselines to enforce compliance settings on devices.

Content Management

- **Distribution Points:** Manage the distribution of content to distribution points, ensuring that software, updates, and operating systems are available to clients.

The **Software Library** tab is essential for IT administrators to efficiently manage software deployment, updates, and compliance across their organization's network.

The **Monitoring** tab in SCCM (System Center Configuration Manager) is essential for tracking and managing the health and performance of your SCCM environment. Here's an overview of its main components:

Key Features of the Monitoring Tab:

1. **Site Hierarchy:**
 - **Purpose:** Provides an overview of your SCCM site hierarchy and interstice links.
 - **Uses:** Monitor the health of each site and the status of interstice replication links.
2. **Client Status:**
 - **Purpose:** Displays the status of all clients in your environment.
 - **Uses:** Review overall statistics for client activity and health checks across the site.
3. **Deployments:**
 - **Purpose:** Track the status of software deployments, including applications, packages, and task sequences.
 - **Uses:** Monitor the success, in-progress, and error states of deployments.
4. **Alerts:**
 - **Purpose:** Manage and view alerts generated by SCCM.
 - **Uses:** Quickly identify and respond to issues that require attention.
5. **Endpoint Protection:**

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- **Purpose:** Monitor the status of endpoint protection policies and malware detections.
 - **Uses:** Ensure that devices are protected and compliant with security policies.
6. **Compliance Settings:**
- **Purpose:** Track the compliance status of configuration items and baselines.
 - **Uses:** Ensure devices meet the required configuration and security standards
7. **Software Updates:**
- **Purpose:** Monitor the deployment and compliance status of software updates.
 - **Uses:** Ensure that devices are up to date with the latest patches and security updates.
8. **Content Status:**
- **Purpose:** Track the distribution status of content to distribution points.
 - **Uses:** Ensure that content is successfully distributed and available to clients

The **Monitoring** tab is crucial for maintaining the overall health and efficiency of your SCCM infrastructure.

To install SCCM (System Center Configuration Manager), there are several prerequisites you need to meet. Here's a summary of the key requirements:

General Requirements

- **Operating System:** Each site system server must use a 64-bit OS. Server Core installations are supported only for the distribution point role.
- **.NET Framework:** Starting from version 2303, SCCM requires Microsoft .NET Framework version 4.8.
- **SQL Server:** SQL Server must be installed and configured correctly. SCCM supports SQL Server Always On availability groups for high availability.

Site System Roles

- **Primary Site and Central Administration Site (CAS):**
 - **Windows Server Roles and Features:** Install necessary roles and features such as IIS, BITS, and Remote Differential Compression.
 - **Permissions:** The user account installing the site must have administrative permissions on the site server and SQL Server.
- **Secondary Site:**
 - **Network Configuration:** Ensure proper network connectivity and bandwidth between the primary site and secondary site.

Hardware Requirements

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- **Processor and Memory:** Follow the processor and RAM requirements for the OS. For example, a primary site server typically requires a multi-core processor and at least 16 GB of RAM.
- **Disk Space:** Ensure sufficient disk space for the site server and SQL Server databases. A minimum of 500 MB is required for the Configuration Manager client cache.

Additional Components

- **ODBC Driver for SQL Server:** Required starting from version 2309.
- **Internet Access:** The site server needs access to the latest setup files from Microsoft, either through direct internet access or a local copy.

Pre-Installation Steps

- **Prerequisite Checker:** Run the Prerequisite Checker to identify and resolve any issues before starting the installation.

Installing SCCM (System Center Configuration Manager) involves several steps to ensure a successful setup. Here's a high-level overview of the process:

1. Prepare the Environment

- **Active Directory:** Extend the Active Directory schema for SCCM and create the System Management container.
- **Service Accounts:** Create necessary service accounts for SCCM and SQL Server.
- **Firewall Configuration:** Ensure the required ports are open for SCCM communication.

2. Install Prerequisites

- **Windows Server Roles and Features:** Install IIS, BITS, RDC, and other necessary roles and features.
- **.NET Framework:** Install .NET Framework 4.8.
- **Windows ADK:** Install Windows Assessment and Deployment Kit (ADK) and Windows PE add-on.
- **SQL Server:** Install SQL Server and configure it for SCCM, including SQL Server Management Studio and Reporting Services.

3. Install SCCM

- **Run the SCCM Setup:** Launch the SCCM setup and follow the wizard.
 - **Select Site Type:** Choose whether to install a primary site, secondary site, or central administration site (CAS).

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- **Configure Site Settings:** Enter site code, site name, and other configuration settings.
- **Database Configuration:** Specify the SQL Server instance and database settings.
- **Configure Service Connection Point:** Set up the service connection point for updates and cloud services.

4. Post-Installation Configuration

- **Configure Site Roles:** Install and configure site system roles such as Management Point, Distribution Point, and Software Update Point.
- **Client Installation:** Deploy the SCCM client to devices in your network.
- **Verify Installation:** Check the status of site components and ensure everything is functioning correctly.

SCCM Client Installation Steps

1. Client Push Installation

- **Setup:** In the SCCM console, go to **Administration > Site Configuration > Sites**. Right-click your site and select **Client Installation Settings > Client Push Installation**.
- **Configuration:** Enable automatic site-wide client push installation and specify the accounts with administrative rights.
- **Deployment:** Right-click a device collection and select **Install Client**.

2. Software Update-Based Installation

- **Setup:** Ensure WSUS is integrated with SCCM and configure the Software Update Point.
- **Configuration:** In the SCCM console, go to **Administration > Site Configuration > Sites**. Right-click your site and select **Configure Site Components > Software Update Point**.
- **Deployment:** Deploy the client software as a software update.

3. Group Policy Installation

- **Setup:** Create a Group Policy Object (GPO) to publish the client installation settings.
- **Configuration:** In the GPO, navigate to **Computer Configuration > Policies > Administrative Templates > System > Configuration Manager** and specify the client installation properties.

4. Manual Installation

- **Setup:** On the target computer, run the command prompt as an administrator.
- **Installation:** Navigate to the folder containing the SCCM client installation files and run `ccmsetup.exe /install`.

Troubleshooting Steps

1. Check Log Files

- **ccmsetup.log:** Located in `C:\Windows\ccmsetup\Logs`, this log file provides details about the client installation process.
- **Client.msi.log:** Located in `C:\Windows\ccmsetup\Logs`, this log file provides details about the MSI installation process.

2. Common Issues and Solutions

- **Firewall and Network Issues:** Ensure that the necessary ports are open and that the client can communicate with the SCCM server.
- **Permissions:** Verify that the installation account has administrative rights on the target computer.
- **WMI Issues:** Repair the WMI repository if it is corrupt.

Common Error Codes

- **0x80070643:** This error often indicates issues with the Windows Installer. Check disk space, repair the Windows Installer, and review the client installation logs.
- **0x87D00215:** Item not found. Ensure that the client installation files are accessible and not corrupted.
- **0x87D00324:** The application was not detected after installation. Verify the detection method and ensure the application is installed correctly.

To set up System Center Configuration Manager (SCCM) servers, you'll need to meet specific Public Key Infrastructure (PKI) certificate requirements. Here are the key points:

Certificate Types and Requirements

1. Server Authentication Certificates:

- **Purpose:** Authenticate servers to clients and encrypt data transfers.
- **Template:** Web Server.
- **Enhanced Key Usage:** Must include Server Authentication (1.3.6.1.5.5.7.3.1).

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- **Subject Name:** Should contain the intranet fully qualified domain name (FQDN).
- **Key Length:** Maximum supported key length is 2,048 bits.
- 2. **Client Authentication Certificates:**
 - **Purpose:** Authenticate clients to the server.
 - **Template:** Workstation Authentication.
 - **Enhanced Key Usage:** Must include Client Authentication (1.3.6.1.5.5.7.3.2).
- 3. **SHA-2 Certificates:**
 - **Requirement:** All internet-facing services should use SHA-2 certificates (e.g., SHA-256, SHA-512) as Windows does not trust SHA-1 certificates.
- 4. **CNG v3 Certificates:**
 - **Support:** Configuration Manager supports Cryptography: Next Generation (CNG) v3 certificates, which can use hardware-based private keys like TPM KSP for PKI client authentication certificates.

Additional Considerations

- **Active Directory Certificate Services:** Required for enrolling client certificates on mobile devices and Mac computers.
- **Enhanced HTTP:** If managing PKI certificates is challenging, you can enable Enhanced HTTP, which uses self-signed certificates along with Microsoft Entra ID for some scenarios.

Installation Steps for SUP/WSUS

1. Install WSUS Role on the Server

1. **Open Server Manager:** Click on "Manage" and select "Add Roles and Features".
2. **Role-based or feature-based installation:** Choose this option and click "Next".
3. **Select Server Roles:** Check "Windows Server Update Services" and proceed.
4. **WSUS Configuration:** Choose the database type (typically SQL Server) and specify the content location.
5. **Complete Installation:** Follow the prompts to complete the installation.

2. Configure WSUS

1. **Post-Installation Tasks:** Open the WSUS console and complete the initial configuration wizard.
2. **Synchronize Updates:** Select the products and classifications you want to synchronize.
3. **Approve Updates:** Manually approve updates for deployment.

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3. Install Software Update Point (SUP) in SCCM

1. **Open SCCM Console:** Navigate to "Administration" > "Site Configuration" > "Servers and Site System Roles".
2. **Add Site System Role:** Right-click the server and select "Add Site System Roles".
3. **Select SUP Role:** Choose "Software Update Point" and configure settings such as ports, proxy, and synchronization schedule.
4. **Complete Configuration:** Follow the prompts to complete the SUP setup.

Troubleshooting Steps and Common Error Codes

1. WSUS Synchronization Issues

- **Error 0x80131509:** This error often indicates a timeout. Try reducing the number of products and classifications selected for synchronization.
- **Error 0x80070643:** This fatal error during installation can be resolved by ensuring all required services are running and the server has internet access.

2. WSUS Connection Failures

- **Verify Services:** Ensure the Update Services service and World Wide Web Publishing Service are running on the WSUS server.
- **Check IIS Logs:** Review logs in `c:\inetpub\logfiles` for errors.

3. General Maintenance

- **Database Cleanup:** Regularly run the WSUS Cleanup Wizard to remove obsolete updates.
- **Reindex Database:** Periodically reindex the WSUS database to improve performance.

To set up a Cloud Management Gateway (CMG) in SCCM, you need to ensure several prerequisites are met:

1. **Azure Subscription:** You need an active Azure subscription to host the CMG.
2. **Certificates:**
 - **Server Authentication Certificate:** This certificate is used by the CMG for HTTPS communication. It can be obtained from a public Certificate Authority (CA) or issued by your internal Public Key Infrastructure (PKI).
 - **Trusted Root Certificate:** This certificate must be trusted by the clients and the CMG.
3. **Azure AD App Registration:**
 - Register the Configuration Manager in Azure AD to allow communication between SCCM and Azure.
4. **Service Connection Point:**

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- Ensure the service connection point in SCCM is in online mode to deploy the CMG in Azure.
5. **Network Requirements:**
 - Ensure internet access for the CMG and clients.
 - Configure necessary network ports for communication.
 6. **Client Configuration:**
 - Clients should be configured to use the CMG for management.
 7. **On-Premises Infrastructure:**
 - A Windows server to host the CMG connection point.

To register SCCM in Azure AD, follow these steps:

1. **Open the SCCM Console:**
 - Go to **Administration > Cloud Services > Azure Services**.
2. **Start the Azure Services Wizard:**
 - Click on **Configure Azure Services**.
 - Select **Cloud Management** and click **Next**.
3. **Create Server and Client Apps:**
 - In the **Azure Services Wizard**, choose to create new server and client apps.
 - Provide a name for the server app (e.g., CMG-ServerApp) and client app (e.g., CMG-ClientApp).
4. **Specify Azure AD Tenant Information:**
 - Enter your **Azure AD tenant name** and **tenant ID**. You can find these in the Azure portal under **Azure Active Directory > Properties**.
5. **Configure App Permissions:**
 - The wizard will automatically configure the necessary permissions for the server and client apps.
6. **Complete the Wizard:**
 - Review the settings and complete the wizard. SCCM will now be registered in Azure AD.

Setting up a Cloud Management Gateway (CMG) in SCCM involves several steps. Here's a detailed guide to help you through the process:

1. **Prerequisites:**

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- Ensure you have the necessary certificates (server authentication certificate, trusted root certificate, etc.).
 - Set up your Azure environment and subscription.
2. **Create the CMG in SCCM:**
 - Open the SCCM console.
 - Navigate to **Administration > Cloud Services > Cloud Management Gateway**.
 - Right-click on **Cloud Management Gateway** and select **Create Cloud Management Gateway**.
 3. **Configure the CMG:**
 - Specify the Azure environment (e.g., AzurePublicCloud).
 - Choose the deployment method (virtual machine scale set is recommended).
 - Configure the primary site for client certificate authentication.
 - Add the CMG connection point site system role.
 4. **Configure Client-Facing Roles:**
 - Configure the management point and software update point for CMG traffic.
 - Set up boundary groups to include the CMG.
 5. **Verify and Monitor:**
 - Check the CMG status in the SCCM console.
 - Ensure clients can communicate with the CMG.

Onboarding Microsoft Defender with SCCM

Here are the detailed steps to onboard Microsoft Defender for Endpoint using System Center Configuration Manager (SCCM):

1. Prerequisites

- **Ensure SCCM is up to date:** Make sure you are using the latest version of SCCM.
- **Endpoint Protection Point Role:** This role must be installed and configured in SCCM.
- **Microsoft Defender for Endpoint Subscription:** Ensure you have the necessary licenses.

2. Enable Endpoint Protection in SCCM

1. **Open SCCM Console:** Navigate to "Administration" > "Overview" > "Site Configuration" > "Sites".
2. **Configure Site Components:** Select "Configure Site Components" > "Software Update Point".

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3. **Enable Endpoint Protection:** Check the box to enable Endpoint Protection.

3. Create Device Collection

1. **Navigate to Assets and Compliance:** Go to "Assets and Compliance" > "Device Collections".
2. **Create New Collection:** Right-click "Device Collections" and select "Create Device Collection".
3. **Define Collection Criteria:** Set criteria to include devices you want to onboard.

4. Configure Microsoft Defender for Endpoint Policies

1. **Create Configuration Item:** Go to "Assets and Compliance" > "Compliance Settings" > "Configuration Items".
2. **Add Configuration Item:** Right-click and select "Create Configuration Item".
3. **Define Settings:** Configure settings for Microsoft Defender Antivirus and other Defender capabilities.

5. Deploy Configuration Baseline

1. **Create Configuration Baseline:** Go to "Assets and Compliance" > "Compliance Settings" > "Configuration Baselines".
2. **Add Configuration Baseline:** Right-click and select "Create Configuration Baseline".
3. **Deploy Baseline:** Deploy the baseline to the device collection created earlier.

6. Monitor Deployment

1. **Check Compliance:** Go to "Monitoring" > "Deployments" to check the compliance status of the deployed baseline.
2. **Review Logs:** Check logs in the SCCM console and on client devices for any errors.

Troubleshooting Common Issues

1. Deployment Failures

- **Error 0x80070643:** This error often indicates issues with the installation of the Endpoint Protection client. Ensure all prerequisites are met and the client can communicate with the SCCM server.

2. Compliance Issues

- **Non-Compliant Devices:** If devices are not compliant, check the configuration item settings and ensure they are correctly applied to the target devices.

Steps to Create Configuration Items and Baselines in SCCM

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1. Create Configuration Items (CIs)

1. **Open SCCM Console:**
 - Navigate to **Assets and Compliance > Compliance Settings > Configuration Items**.
2. **Create New Configuration Item:**
 - Right-click on **Configuration Items** and select **Create Configuration Item**.
 - Enter a **name** and **description** for the CI.
3. **Define Settings:**
 - Choose the **settings type** (e.g., Windows, Mac OS, etc.).
 - Add **detection methods**, **compliance rules**, and **supported platforms**.
4. **Configure Compliance Rules:**
 - Define rules to check for compliance (e.g., registry settings, file existence, etc.).
5. **Complete the Wizard:**
 - Review the settings and complete the wizard to create the CI.

2. Create Configuration Baselines (CBs)

1. **Navigate to Configuration Baselines:**
 - Go to **Assets and Compliance > Compliance Settings > Configuration Baselines**.
2. **Create New Baseline:**
 - Right-click on **Configuration Baselines** and select **Create Configuration Baseline**.
 - Enter a **name** and **description** for the baseline.
3. **Add Configuration Items:**
 - Click **Add** to include the previously created CIs or other baselines.
 - You can also include **software updates** and other configuration baselines.
4. **Set Compliance Evaluation:**
 - Specify whether the items are **required** or **optional** for compliance evaluation.
5. **Complete the Wizard:**
 - Review the settings and complete the wizard to create the baseline.

3. Deploy Configuration Baselines

1. **Deploy Baseline:**
 - Right-click on the created baseline and select **Deploy**.
 - Choose the **target collection** (e.g., a specific device collection).

2. Monitor Compliance:

- Go to **Monitoring** > **Deployments** to check the compliance status of the deployed baseline.

Additional Tips

- **Regular Updates:** Keep your configuration items and baselines updated to reflect any changes in your environment.
- **Testing:** Test new configuration items and baselines in a controlled environment before deploying them widely.

Steps to Configure Software Metering in SCCM

1. Enable Software Metering in Client Settings

1. Open SCCM Console:

- Navigate to **Administration** > **Client Settings** > **Default Client Settings**.

2. Configure Software Metering:

- In the **Default Settings** dialog box, click **Software Metering**.
- Set **Enable software metering on clients** to **True**.
- Configure the **Schedule data collection** interval (default is every 7 days).

3. Deploy Client Settings:

- Deploy these settings to the desired device collections.

2. Create Software Metering Rules

1. Navigate to Software Metering:

- Go to **Assets and Compliance** > **Software Metering**.

2. Create New Rule:

- On the Home tab, click **Create Software Metering Rule**.
- In the wizard, specify the following:
 - **Name:** Enter a name for the rule.
 - **File Name:** Specify the executable file name (e.g., `msedge.exe` for Microsoft Edge).
 - **Original File Name:** Enter the original file name if known.
 - **Version:** Specify the version if applicable (use * for all versions).
 - **Site:** Select the site where the rule will apply.

3. Complete the Wizard:

- Review the settings and complete the wizard to create the rule.

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3. Monitor Software Metering

1. Run Reports:

- Go to **Monitoring > Reporting > Reports > Software Metering**.
- Select a report, such as **Computers that have run a specific metered software program** and run it to view usage data.

2. Review Data:

- Analyze the reports to understand software usage patterns and make informed decisions about software licensing and deployment.

Steps to Configure Power Management in SCCM

1. Enable Power Management in Client Settings

1. Open SCCM Console:

- Navigate to **Administration > Client Settings > Default Client Settings**.

2. Configure Power Management:

- In the **Default Settings** dialog box, click **Power Management**.
- Set **Allow power management of devices** to **True**.
- Configure additional settings as needed, such as **Peak and Non-Peak Hours**.

3. Deploy Client Settings:

- Deploy these settings to the desired device collections.

2. Create and Apply Power Plans

1. Navigate to Device Collections:

- Go to **Assets and Compliance > Device Collections**.

2. Select Collection:

- Choose the device collection to which you want to apply power management settings.

3. Configure Power Management Settings:

- Right-click the collection and select **Properties**.
- Switch to the **Power Management** tab.
- Select **Specify power management settings for this collection**.
- Configure settings such as **Start and End time for peak hours**, **Sleep settings**, and **Wakeup time**.

4. Apply Power Plan:

- Save the settings to apply the power plan to the selected device collection.

3. Monitor and Report Power Management

1. Run Power Management Reports:

- Go to **Monitoring > Reporting > Reports > Power Management**.
- Select and run reports like **Computers with Multiple Power Plans** to analyze power consumption and settings.

2. Review Data:

- Use the reports to identify and resolve any power management conflicts or issues.

Types of Application Deployments in SCCM

SCCM supports various types of application deployments, each suited for different scenarios. Here are the main types:

1. Windows Installer (MSI)

- **Purpose:** Deploy applications packaged as MSI files.
- **Example:** Deploying Microsoft Office.

2. Script Installer

- **Purpose:** Use scripts to install applications.
- **Example:** Custom scripts for installing software that doesn't have an MSI package.

3. Microsoft App-V

- **Purpose:** Deploy virtual applications using Microsoft Application Virtualization.
- **Example:** Deploying virtualized versions of applications to reduce conflicts.

4. Windows Store Apps

- **Purpose:** Deploy applications from the Microsoft Store.
- **Example:** Deploying apps like Microsoft To-Do.

5. Mac OS X

- **Purpose:** Deploy applications to Mac devices.
- **Example:** Deploying software like Microsoft Office for Mac.

6. Task Sequence

- **Purpose:** Deploy complex applications that require multiple steps.
- **Example:** Deploying an application that requires pre-requisites and post-installation configurations.

Steps to Create Application Deployments in SCCM

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1. Prepare the Application

1. **Gather Installation Files:** Ensure you have the necessary installation files and any required scripts or dependencies.

2. Create the Application in SCCM

1. **Open SCCM Console:**
 - Navigate to **Software Library > Application Management > Applications**.
2. **Create New Application:**
 - Click on **Create Application** in the ribbon.
 - Choose the type of application (e.g., Windows Installer, Script Installer).
 - Specify the location of the installation files.
3. **General Information:**
 - Enter the application name, publisher, and version.
 - Optionally, add a description and administrative categories.
4. **Deployment Types:**
 - Add a deployment type by specifying the installation command, detection method, and requirements.
 - Configure user experience settings, such as installation behavior and restart options.
5. **Content:**
 - Specify the content location and distribution points.
6. **Requirements:**
 - Define any requirements for the application, such as operating system versions or hardware specifications.
7. **Dependencies:**
 - Add any dependencies that must be installed before the application.

3. Deploy the Application

1. **Select the Application:**
 - Go to **Software Library > Application Management > Applications**.
 - Select the application you created.
2. **Deploy:**
 - Click on **Deploy** in the ribbon.
 - Choose the target collection (e.g., a specific device or user collection).
3. **Deployment Settings:**

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- Configure the deployment settings, such as action (Install/Uninstall) and purpose (Available/Required).
- 4. **Scheduling:**
 - Set the schedule for when the application should be deployed.
- 5. **User Experience:**
 - Configure user notifications and installation behavior.
- 6. **Alerts:**
 - Set up alerts to monitor the deployment status.

4. Monitor the Deployment

1. **Check Deployment Status:**
 - Go to **Monitoring > Deployments**.
 - Review the status of the deployment and troubleshoot any issues.

Types of Package Deployments in SCCM

SCCM supports various types of package deployments, each suited for different scenarios. Here are the main types:

1. **Standard Package Deployment**
 - **Purpose:** Deploy software that doesn't require complex installation logic.
 - **Example:** Deploying a simple application or script.
2. **Software Update Package**
 - **Purpose:** Deploy software updates to client machines.
 - **Example:** Deploying Windows updates or third-party software patches.
3. **Driver Package**
 - **Purpose:** Deploy drivers to client machines.
 - **Example:** Deploying hardware drivers during OS deployment.
4. **Operating System Image Package**
 - **Purpose:** Deploy operating system images.
 - **Example:** Deploying a Windows 10 image to new machines.
5. **Boot Image Package**
 - **Purpose:** Deploy boot images used in OS deployment.
 - **Example:** Deploying a WinPE boot image for PXE boot.

Steps to Create Package Deployments in SCCM

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1. Create a Standard Package

1. **Open SCCM Console:**
 - Navigate to **Software Library > Application Management > Packages**.
2. **Create New Package:**
 - Click on **Create Package** in the ribbon.
 - Enter the **name**, **description**, and **manufacturer** (optional).
 - Specify whether the package contains source files and provide the source folder location.
3. **Create Program:**
 - After creating the package, right-click on it and select **Create Program**.
 - Enter the **name** and **command line** for the program (e.g., `setup.exe /silent`).
 - Configure additional settings like **run mode** (e.g., normal, hidden) and **requirements**.
4. **Distribute Content:**
 - Right-click the package and select **Distribute Content**.
 - Choose the **distribution points** where the package will be available.
5. **Deploy Package:**
 - Right-click the package and select **Deploy**.
 - Choose the **target collection** (e.g., a specific device collection).
 - Configure deployment settings such as **action** (Install/Uninstall) and **purpose** (Available/Required).

2. Create a Software Update Package

1. **Open SCCM Console:**
 - Navigate to **Software Library > Software Updates > All Software Updates**.
2. **Select Updates:**
 - Select the updates you want to include in the package.
 - Right-click and choose **Create Software Update Group**.
3. **Download Updates:**
 - Right-click the update group and select **Download**.
 - Specify the **download location** and **distribution points**.
4. **Deploy Update Group:**
 - Right-click the update group and select **Deploy**.
 - Choose the **target collection** and configure deployment settings.

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3. Create a Driver Package

1. **Open SCCM Console:**
 - Navigate to **Software Library > Operating Systems > Driver Packages**.
2. **Create New Driver Package:**
 - Click on **Create Driver Package** in the ribbon.
 - Enter the **name** and **path** where the drivers are stored.
3. **Add Drivers:**
 - Import drivers into SCCM and add them to the driver package.
4. **Distribute Content:**
 - Distribute the driver package to the desired distribution points.
5. **Deploy Driver Package:**
 - Use task sequences to deploy the driver package during OS deployment.

Creating and Deploying a Software Update Group (SUG) in SCCM

1. Create a Software Update Group (SUG)

1. **Open SCCM Console:**
 - Navigate to **Software Library > Software Updates > All Software Updates**.
2. **Select Updates:**
 - Use the search and filter options to find the updates you want to include.
 - Select the updates by checking the boxes next to them.
3. **Create Software Update Group:**
 - On the **Home** tab, in the **Update** group, click **Create Software Update Group**.
 - Enter a **name** and **description** for the update group.
 - Click **Create** to finalize the group.

2. Deploy the Software Update Group

1. **Select the Software Update Group:**
 - Go to **Software Library > Software Updates > Software Update Groups**.
 - Select the update group you created.
2. **Deploy the Update Group:**
 - Right-click the update group and select **Deploy**.
 - In the **Deploy Software Updates Wizard**, specify the following:
 - **Collection:** Choose the target collection (e.g., a specific device collection).

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- **Deployment Settings:** Configure settings such as **action** (Install/Uninstall) and **purpose** (Available/Required).
 - **Scheduling:** Set the schedule for when the updates should be deployed.
 - **User Experience:** Configure user notifications and installation behavior.
 - **Alerts:** Set up alerts to monitor the deployment status.
3. **Distribute Content:**
- Ensure the update content is distributed to the appropriate distribution points.
 - Go to **Monitoring > Distribution Status > Content Status** to check the distribution status.
4. **Monitor Deployment:**
- Go to **Monitoring > Deployments** to review the status of the deployment.
 - Use reports and logs to troubleshoot any issues.

Creating an Automatic Deployment Rule (ADR) in SCCM

1. Create an Automatic Deployment Rule

1. **Open SCCM Console:**
 - Navigate to **Software Library > Software Updates > Automatic Deployment Rules**.
2. **Create New ADR:**
 - Click on **Create Automatic Deployment Rule** in the ribbon.
 - Enter a **name** and **description** for the ADR.
 - Choose a **template** (e.g., Patch Tuesday) if applicable.
3. **Specify Collection:**
 - Click **Browse** and select the target **collection** for the update deployment.
4. **Configure Settings:**
 - Choose whether to create a new Software Update Group or add to an existing one.
 - Enable the deployment after the rule runs.
5. **Deployment Settings:**
 - Set the detail level to **Only success and error messages**.
 - Choose to automatically deploy all software updates found by the rule and approve any license agreements.
6. **Specify Criteria:**
 - Define the criteria for updates (e.g., date released, language, product, classification).

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7. Evaluation Schedule:

- Configure when the ADR should be evaluated (e.g., monthly, weekly).

8. Deployment Schedule:

- Set the schedule for when the updates should be deployed to clients.

9. User Experience:

- Configure user notifications and installation behavior.

10. Alerts:

- Set up alerts to monitor the deployment status.

11. Download Settings:

- Specify the download settings and distribution points.

12. Complete the Wizard:

- Review the settings and complete the wizard to create the ADR.

2. Troubleshooting Common ADR Issues

1. Updates Not Downloading:

- **Error 0x80070050:** This error indicates a problem with temporary files. Clear the `C:\Windows\Temp` directory and rerun the ADR.
- **Error 0x87D20417:** This error can be related to proxy settings or incorrect paths. Check the `ruleengine.log` and `patchdownloader.log` for more details.

2. Rule Evaluation Failures:

- Ensure that the site has completed software updates synchronization before creating the ADR.
- Verify that the criteria specified in the ADR match the updates available.

3. Content Distribution Issues:

- Check the distribution point status and ensure that the content is properly distributed.
- Review the `distmgr.log` for any errors related to content distribution.

4. General Maintenance:

- Regularly review and clean up old software update groups and deployments to avoid clutter.
- Keep SCCM and WSUS up to date with the latest patches and updates.

Creating an Operating System Deployment (OSD) in SCCM

1. Prepare the Environment

1. Install Windows ADK:

- Ensure the Windows Assessment and Deployment Kit (ADK) is installed on your SCCM server.
- Install the Windows PE add-on for ADK.

2. Distribute Boot Images:

- Navigate to **Software Library > Operating Systems > Boot Images**.
- Right-click the boot image and select **Distribute Content** to distribute it to distribution points.

2. Import Operating System Image

1. Add OS Image:

- Go to **Software Library > Operating Systems > Operating System Images**.
- Right-click and select **Add Operating System Image**.
- Specify the path to the `install.wim` file from your Windows installation media.

2. Distribute OS Image:

- Right-click the imported OS image and select **Distribute Content**.
- Choose the distribution points to distribute the OS image.

3. Create a Task Sequence

1. Create Task Sequence:

- Navigate to **Software Library > Operating Systems > Task Sequences**.
- Click **Create Task Sequence** and choose **Install an existing image package**.

2. Configure Task Sequence:

- Provide a name and boot image for the task sequence.
- Select the OS image you imported earlier.
- Configure additional steps such as applying drivers, installing software, and configuring settings.

3. Distribute Task Sequence:

- Right-click the task sequence and select **Distribute Content**.
- Ensure all referenced content is distributed to the distribution points.

4. Deploy the Task Sequence

1. Deploy Task Sequence:

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- Right-click the task sequence and select **Deploy**.
- Choose the target collection (e.g., a specific device collection).
- Configure deployment settings such as **purpose** (Available/Required) and **schedule**.

2. Monitor Deployment:

- Go to **Monitoring** > **Deployments** to check the status of the deployment.

Troubleshooting Common OSD Issues

1. Task Sequence Errors

- **Error 0x80070002:** This error often indicates missing files. Ensure all content is properly distributed to the distribution points.
- **Error 0x80004005:** This generic error can be due to various issues. Check the `smsts.log` file for more details.

2. Log Files for Troubleshooting

- **SMSTS.log:** The primary log file for task sequence issues. Locations vary depending on the deployment phase:
 - **WinPE (before HDD format):** `x:\windows\temp\smstslog\smsts.log`
 - **WinPE (after HDD format):** `c:\_SMSTaskSequence\Logs\Smstslog\smsts.log`
 - **Full OS (before SCCM agent installed):** `c:\_SMSTaskSequence\Logs\Smstslog\smsts.log`
 - **Full OS (after SCCM agent installed):** `%windir%\ccm\logs\Smstslog\smsts.log`

3. Common Issues and Fixes

- **Content Distribution Failures:**
 - Ensure all referenced content is distributed to the correct distribution points.
 - Check the `distmgr.log` for any errors related to content distribution.
- **Driver Issues:**
 - Ensure the correct drivers are included in the task sequence.
 - Use the `drivercatalog.log` to troubleshoot driver-related issues.

Key Components of Microsoft Edge Management

1. Microsoft Edge Management Service:

- **Platform:** Available in the Microsoft 365 admin center.

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- **Purpose:** Allows admins to configure browser settings, manage extensions, and apply policies to users' browsers.
 - **Configuration Profiles:** Admins can create and assign configuration profiles to different groups, ensuring consistent settings across devices.
2. **Configuration Manager (SCCM):**
 - **Deployment:** Admins can deploy Microsoft Edge using SCCM, choosing from Beta, Dev, or Stable channels.
 - **Updates:** Manage updates for Microsoft Edge, either allowing automatic updates or controlling them through SCCM.
 3. **Group Policy and Microsoft Entra ID:**
 - **Policy Settings:** Configure Microsoft Edge settings using Group Policy or Microsoft Entra ID (formerly Azure AD) for centralized management.
 - **Priority Handling:** When multiple policies apply, the highest priority setting is used.

Steps to Manage Microsoft Edge

1. Access Microsoft Edge Management Service

1. **Sign In:**
 - Go to the Microsoft 365 admin center and sign in with your admin credentials.
2. **Navigate to Edge Settings:**
 - In the left navigation bar, go to **Settings > Microsoft Edge**.

2. Create Configuration Profiles

1. **Add a Profile:**
 - Under the **Configuration profiles** section, select **Add a profile**.
 - Enter a **profile name** and **description**, then click **Add**.
2. **Configure Policies:**
 - After creating the profile, configure the desired policies and extensions.
 - Assign the profile to Microsoft Entra groups as needed.

3. Deploy Microsoft Edge Using SCCM

1. **Create Edge Application:**
 - In SCCM, navigate to **Software Library > Application Management > Applications**.
 - Create a new application for Microsoft Edge, specifying the channel and version.
2. **Distribute Content:**

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- Distribute the application content to the appropriate distribution points.
3. **Deploy Application:**
 - Deploy the application to the target device collection, configuring deployment settings and schedules.

Troubleshooting Common Issues

1. **Policy Conflicts:**
 - Ensure that there are no conflicting policies across different configuration profiles. The highest priority policy will be applied if conflicts exist.
2. **Update Failures:**
 - Verify that the devices have access to the necessary endpoints for downloading updates.
 - Check the SCCM logs for any errors related to deployment or updates.

Reporting is the process of collecting, processing, storing, and presenting information within an organization. It involves creating documents or dashboards that communicate essential information, such as project status, sales numbers, or budget reports.

Reporting helps in making informed business decisions, creating transparency, and facilitating knowledge transfer within the company.

To **configure reporting**, especially in a system like SQL Server Reporting Services (SSRS), you can follow these general steps.

1. **Install SQL Server Reporting Services (SSRS):**
 - Download SSRS from the Microsoft website.
 - Run the installer and choose the installation location.
2. **Configure the Report Server:**
 - Open the Report Server Configuration Manager.
 - Create a new report server database and configure the database connection.
 - Set up the web service URL and the web portal URL for hosting and accessing reports.
3. **Set Up Security and Permissions:**
 - Use the web portal to grant permissions to users or groups.
 - Configure security settings to control access to reports.
4. **Optional Configurations:**
 - Set up email settings for report distribution.
 - Backup encryption keys and define file share accounts if needed.

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System Center Configuration Manager (SCCM), the **site system status** provides an overview of the health and operational status of site system servers and components within your SCCM hierarchy. This status helps administrators monitor and manage the infrastructure effectively, ensuring that all components are functioning correctly.

How to Troubleshoot Site System Status in SCCM

1. **Check Status Messages:**
 - Navigate to the **Monitoring** workspace in the SCCM console.
 - Go to **System Status > Site Status** or **Component Status**.
 - Review the status messages for any errors or warnings. These messages provide detailed information about issues.
2. **Review Logs:**
 - Examine relevant log files located in the `Logs` folder on the site server. Key logs include `sitelog.log`, `componentstatus.log`, and `statmgr.log`.
 - Use tools like **CMTrace** to read and filter log files efficiently.
3. **Monitor Performance Metrics:**
 - Check the performance of site system servers using tools like **Performance Monitor**.
 - Look for high CPU, memory, or disk usage that might indicate underlying issues.
4. **Verify Network Connectivity:**
 - Ensure that all site system servers can communicate with each other.
 - Check for any network-related issues that might be affecting site system operations.
5. **Update and Patch:**
 - Make sure that all site system servers are running the latest updates and patches.
 - Regularly update SCCM to the latest version to benefit from fixes and improvements.
6. **Check Disk Space:**
 - Ensure that there is sufficient disk space on site system servers.
 - Low disk space can cause various issues, including failed deployments and status message processing errors.
7. **Restart Services:**
 - Sometimes, restarting SCCM services on the site system servers can resolve issues.
 - Use the **Services** console or command line to restart services like `SMS_EXECUTIVE` and `SMS_SITE_COMPONENT_MANAGER`.

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Monitoring deployments in **System Center Configuration Manager (SCCM)** is crucial to ensure that applications, updates, and operating systems are successfully deployed across your environment. Here are the steps to monitor deployments in SCCM:

Monitoring Application Deployments

1. Using the SCCM Console:

- Go to the **Monitoring** workspace.
- Click on **Deployments** to see a list of all deployments.
- Select the deployment you want to monitor and click **View Status** on the Home tab.

2. Compliance States:

- SCCM tracks the state of application deployments using compliance states such as **Success**, **In Progress**, **Unknown**, **Requirements Not Met**, and **Error**.
- You can view detailed information about each compliance state to identify any issues.

3. Reports:

- SCCM includes built-in reports that provide detailed information about application deployments.
- Navigate to **Monitoring > Reporting > Reports** and select the relevant report to view deployment details.

Monitoring Operating System Deployments

1. Task Sequence Deployment Status:

- In the **Monitoring** workspace, click **Deployments**.
- Select the task sequence deployment you want to monitor and click **View Status**.

2. Alerts:

- Configure alerts in the task sequence deployment settings to notify you when compliance levels fall below a certain percentage.
- Review these alerts in the **Operating Systems** node in the **Software Library** workspace or the **Alerts** node in the **Monitoring** workspace.

3. Reports:

- SCCM provides predefined reports for operating system deployments, which can be accessed under **Monitoring > Reporting > Reports**.

Monitoring client status in **System Center Configuration Manager (SCCM)** is essential to ensure that all client devices are healthy and communicating properly with the SCCM server. Here's how you can monitor client status in SCCM:

Steps to Monitor Client Status

1. **Access the Monitoring Workspace:**
 - Open the SCCM console.
 - Navigate to the **Monitoring** workspace.
 - Select **Client Status** from the left-hand pane.
2. **View Client Status:**
 - In the **Client Status** node, you can see an overview of the status of all clients.
 - This includes information on client online status, client activity, and client checks.
3. **Configure Client Status Settings:**
 - Go to the **Client Status Settings** on the Home tab.
 - Configure settings such as the number of days since the client requested policy, sent a heartbeat message, or sent hardware inventory.
 - These settings help determine when a client is considered inactive.
4. **Monitor Individual Clients:**
 - In the **Assets and Compliance** workspace, select **Devices**.
 - Right-click on a device and select **Start > Resource Explorer** to view detailed information about the client.
5. **Use Reports:**
 - SCCM provides built-in reports for client status monitoring.
 - Navigate to **Monitoring > Reporting > Reports** and select the relevant report to view detailed client status information.
6. **Set Up Alerts:**
 - Configure alerts to notify you if client activity falls below a specified threshold.
 - This can be done in the **Client Status Settings**.

Key Metrics to Monitor

- **Client Online Status:** Indicates if the client is currently connected to its assigned management point.
- **Client Activity:** Shows if the client has communicated with SCCM in the past seven days.
- **Client Check:** Periodic evaluations run by the SCCM client to check and remediate issues on the device.

Monitoring distribution status in **System Center Configuration Manager (SCCM)** ensures that content such as applications, packages, and updates are successfully distributed to distribution points. Here's how you can monitor distribution status in SCCM:

Steps to Monitor Distribution Status

1. **Using the SCCM Console:**
 - Navigate to the **Monitoring** workspace.
 - Expand **Distribution Status** and select **Content Status**.
 - This node provides an overview of the distribution status for all content. You can see whether the content distribution is successful, in progress, or has failed.
2. **View Content Distribution Graph:**
 - Starting from SCCM version 2203, you can visualize content distribution status in a graphical format.
 - Right-click on a content item (e.g., application or package) and select **View Content Distribution**. This action displays a graph showing the distribution point type, distribution state, and associated status messages.
3. **Check Distribution Point Status:**
 - Go to **Monitoring > System Status > Site Status**.
 - Sort by **Site System Role** to list only distribution points.
 - Ensure that distribution points have sufficient disk space and are not reporting any errors.
4. **Review Logs:**
 - Examine relevant log files such as `distmgr.log` and `pkgxfermgr.log` on the site server.
 - Use tools like **CMTrace** to read and filter log files efficiently.
5. **Use Reports:**
 - SCCM includes built-in reports for monitoring distribution status.
 - Navigate to **Monitoring > Reporting > Reports** and select the relevant report to view detailed distribution status information.

Key Metrics to Monitor

- **Distribution Point Status:** Ensure all distribution points are healthy and have sufficient resources.
- **Content Distribution Progress:** Track the progress of content distribution to identify any bottlenecks or failures.
- **Error Messages:** Pay attention to any error messages in the logs or status messages to troubleshoot issues promptly.

Monitoring the **Software Update Point (SUP) synchronization status** in **System Center Configuration Manager (SCCM)** is crucial to ensure that your software updates are being synchronized correctly. Here are the steps to monitor the SUP sync status:

Steps to Monitor SUP Sync Status

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1. Using the SCCM Console:

- Open the SCCM console.
- Navigate to the **Monitoring** workspace.
- Select **Software Update Point Synchronization Status**.
- This view provides details such as the last synchronization attempt, synchronization status, and any error codes.

2. Reviewing Logs:

- Check the `wsyncmgr.log` file on the site server. This log file records the synchronization process and any issues that occur.
- Use tools like **CMTrace** to read and filter log files efficiently.

3. Using Reports:

- SCCM includes built-in reports for monitoring software update synchronization.
- Navigate to **Monitoring > Reporting > Reports** and select the relevant report to view detailed synchronization status information.

Key Metrics to Monitor

- **Last Synchronization Attempt:** Indicates the date and time of the last synchronization attempt.
- **Synchronization Status:** Shows whether the synchronization was successful or if there were any errors.
- **Error Codes:** Provides specific error codes that can help diagnose issues.

By following these steps, you can effectively monitor the SUP synchronization status in SCCM and ensure that your software updates are up-to-date and properly synchronized.

Updating and servicing in **System Center Configuration Manager (SCCM)** involves keeping your SCCM infrastructure up to date with the latest features, fixes, and improvements. Here are the steps to perform updates and servicing, along with troubleshooting tips:

Steps to Perform Updates and Servicing

1. Check for Updates:

- Open the SCCM console.
- Navigate to **Administration > Updates and Servicing**.
- The console will display available updates. If no updates are visible, click **Check for Updates** in the ribbon.

2. Download Updates:

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- Select the update you want to install.
 - Click **Download** in the ribbon. The update will be downloaded to the site server.
3. **Install Updates:**
 - Once the update is downloaded, select it and click **Install Update Pack**.
 - Follow the wizard to complete the installation. You can choose to validate the update in a pre-production collection before deploying it to the entire hierarchy.
 4. **Monitor the Installation:**
 - Go to **Monitoring > Updates and Servicing Status** to monitor the progress of the update installation.
 - Review the status messages and logs to ensure the update is applied successfully.

Troubleshooting Update and Servicing Issues

1. **Check Log Files:**
 - Review the `dmpdownloader.log`, `cmupdate.log`, and `hman.log` files on the site server for any errors during the update process.
 - Use **CMTrace** to read and filter log files efficiently.
2. **Verify Prerequisites:**
 - Ensure that all prerequisites for the update are met. This includes having the necessary permissions and ensuring that the site server has sufficient resources.
3. **Restart Services:**
 - Sometimes, restarting SCCM services can resolve issues. Use the **Services** console or command line to restart services like `SMS_EXECUTIVE` and `SMS_SITE_COMPONENT_MANAGER`.
4. **Check Internet Connectivity:**
 - Ensure that the site server has a stable internet connection to download updates.
 - If using a proxy, verify that the proxy settings are correctly configured.
5. **Run the Service Connection Tool:**
 - If your service connection point is in offline mode, use the **Service Connection Tool** to manually sync with the Microsoft cloud.

Configuring a hierarchy in **System Center Configuration Manager (SCCM)** involves setting up a structured environment to manage devices and users efficiently. Here are the steps to configure a hierarchy and how to dive deeper into its components:

Steps to Configure SCCM Hierarchy

1. Plan Your Hierarchy:

- Determine the type of hierarchy you need: a **stand-alone primary site** for smaller environments or a **central administration site (CAS)** with multiple primary sites for larger, distributed environments.
- Consider factors like the number of clients, geographic distribution, and network bandwidth.

2. Install the Central Administration Site (CAS) (if needed):

- The CAS is the top-level site in a hierarchy and does not manage clients directly but provides a central point for administration.
- Install the CAS first if you plan to have multiple primary sites.

3. Install Primary Sites:

- Primary sites manage clients directly and can be installed as stand-alone or as child sites under a CAS.
- Ensure each primary site has sufficient resources and is correctly configured to handle the expected client load.

4. Install Secondary Sites (if needed):

- Secondary sites extend the capabilities of primary sites, typically used to manage clients in remote locations with limited network bandwidth.
- Secondary sites report to a primary site and help reduce network traffic.

5. Configure Site System Roles:

- Assign roles like **Management Point (MP)**, **Distribution Point (DP)**, **Software Update Point (SUP)**, and **Reporting Services Point (RSP)** to appropriate servers.
- Ensure each role is configured correctly to support the hierarchy.

6. Set Up Boundaries and Boundary Groups:

- Define boundaries based on IP subnets, Active Directory sites, or IPv6 prefixes to manage client assignments and content location.
- Group boundaries into boundary groups to associate them with site system roles.

7. Configure Replication:

- Set up SQL-based replication for global data and file-based replication for site-specific data.
- Monitor replication status to ensure data consistency across the hierarchy.

Deep Dive into SCCM Hierarchy Configuration

1. Replication in SCCM:

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- Understand the two types of replications: **SQL-based replication** for global data and **file-based replication** for site-specific data.
- Key logs to monitor replication include `rcmctrl.log`, `replmgr.log`, `sender.log`, and `despooler.log`.

2. Site System Roles:

- Each role has specific functions and requirements. For example, the **Management Point** handles client communication, while the **Distribution Point** stores content for client access.
- Properly configuring and distributing these roles is crucial for efficient hierarchy operation.

3. Monitoring and Troubleshooting:

- Use the **Monitoring** workspace in the SCCM console to track the status of sites, components, and replication.
- Regularly review logs and set up alerts to proactively manage issues.

By following these steps and diving deeper into the components of SCCM hierarchy configuration, you can effectively manage and scale your SCCM deployment.

Configuring and troubleshooting sites, servers, and site system roles can be quite detailed. Here's a general guide to help you get started:

Site Configuration Steps

1. Plan Your Site Hierarchy:

- Determine the number of sites needed (central administration site, primary sites, secondary sites).
- Plan the placement of site system roles based on your network topology and bandwidth.

2. Install the Site:

- Use the Configuration Manager setup wizard to install the site.
- Configure the site server and database server.

3. Configure Site Boundaries:

- Define boundaries to manage devices within your network.
- Create boundary groups to associate boundaries with site system roles.

Server Configuration Steps

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1. **Prepare the Server:**
 - Ensure the server meets hardware and software requirements.
 - Install necessary prerequisites (e.g., .NET Framework, IIS).
2. **Install Site System Roles:**
 - In the Configuration Manager console, go to the Administration workspace.
 - Select the Servers and Site System Roles node.
 - Add site system roles to the server (e.g., management point, distribution point)

Site System Roles Configuration

1. **Management Point:**
 - Install the role on a server to manage client communications.
 - Ensure the server has sufficient resources and network connectivity.
2. **Distribution Point:**
 - Install the role to distribute content to clients.
 - Configure content libraries and distribution settings.
3. **Software Update Point:**
 - Install the role to manage software updates.
 - Synchronize with WSUS and configure update settings.

Troubleshooting Steps

1. **Verify Configuration:**
 - Check that all roles are installed correctly.
 - Ensure that servers meet all prerequisites.
2. **Network Connectivity:**
 - Verify IP address configuration and DNS settings.
 - Check firewall rules and routing tables.
3. **Logs and Monitoring:**
 - Review Configuration Manager logs for errors.
 - Use monitoring tools to check the health of site system roles.
4. **Client Communication:**
 - Ensure clients can communicate with management points.
 - Check client logs for any communication issues.

Configuring client settings in Configuration Manager involves managing both default and custom settings. Here's a guide to help you:

Default Client Settings

1. **Access Default Settings:**
 - In the Configuration Manager console, go to the **Administration** workspace.
 - Select the **Client Settings** node.
 - Choose **Default Client Settings** and click **Properties**.
2. **Configure Settings:**
 - Adjust settings for various categories like **Client Policy, Software Updates, Compliance Settings**, etc.
 - These settings apply to all clients in the hierarchy unless overridden by custom settings.

Custom Client Settings

1. **Create Custom Settings:**
 - In the **Administration** workspace, select the **Client Settings** node.
 - Click on **Create Custom Client Settings** and choose either **Device Settings** or **User Settings**.
2. **Specify Settings:**
 - Provide a unique name and description for the custom settings.
 - Select the settings categories you want to configure (e.g., **Power Management, Remote Tools**).
3. **Deploy Custom Settings:**
 - After configuring, select the custom client setting you created.
 - Click **Deploy** and choose the target collection (e.g., specific devices or users).
 - Custom settings will override default settings for the targeted collection.

Tips for Managing Client Settings

- **Order of Application:** If multiple custom settings are deployed to the same collection, they are applied based on their order number. Lower order numbers take precedence.
- **Policy Retrieval:** Clients will apply new settings during their next policy retrieval cycle. You can manually initiate this process if needed.

Configuring security roles, custom roles, and administrative users in Configuration Manager involves several steps. Here's a guide to help you:

Security Roles Configuration

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1. Access Security Roles:

- In the Configuration Manager console, go to the **Administration** workspace.
- Select **Security**, then **Security Roles**.

2. Create Custom Security Roles:

- Select an existing security role to use as a template.
- Click **Copy** to create a new custom role.
- In the **Copy Security Role** wizard, specify a name and description for the new role.
- Adjust permissions as needed by expanding each object type and setting permissions to **Yes** or **No**.

Custom Roles Configuration

1. Define Custom Roles:

- Custom roles can be tailored to specific job functions.
- Combine necessary permissions from different built-in roles to create a custom role that fits your needs.

2. Assign Custom Roles:

- After creating a custom role, assign it to administrative users.
- Ensure that users have only the permissions they need to perform their tasks.

Administrative Users Configuration

1. Add Administrative Users:

- In the **Administration** workspace, select **Security**, then **Administrative Users**.
- Click **Add User or Group** to add a new administrative user.
- Specify the user or group and assign the appropriate security roles.

2. Configure User Permissions:

- Assign security roles to users based on their job functions.
- Use security scopes and collections to define the objects and resources users can manage.

Tips for Managing Security and Administrative Users

- **Least Privilege Principle:** Always assign the minimum permissions necessary for users to perform their tasks.
- **Regular Audits:** Periodically review and audit security roles and user permissions to ensure compliance and security.
- **Documentation:** Keep detailed documentation of all custom roles and user assignments for reference and troubleshooting.

Migrating in System Center Configuration Manager (SCCM) involves several detailed steps. Here's a guide to help you with the migration process and troubleshooting:

Migration Steps

1. **Plan for Migration:**
 - Understand your current and target hierarchies.
 - Ensure all prerequisites are met, including hardware and software requirements.
2. **Set Up the Destination Hierarchy:**
 - Install and configure the new SCCM environment.
 - Set up the necessary site system roles and configurations.
3. **Create Migration Jobs:**
 - In the SCCM console, go to the **Administration** workspace.
 - Select **Migration**, then **Migration Jobs**.
 - Use the **Create Migration Job** wizard to set up jobs for migrating collections, objects, and content.
4. **Run Migration Jobs:**
 - Execute the migration jobs to transfer data from the source to the destination hierarchy.
 - Monitor the migration process in the **Migration** workspace.
5. **Migrate Clients:**
 - Reassign clients to the new hierarchy.
 - Ensure clients are properly configured and communicating with the new site.
6. **Post-Migration Tasks:**
 - Validate the migration by checking data integrity and system functionality.
 - Optimize the new environment for performance and efficiency.

Troubleshooting Steps

1. **Verify Configuration:**
 - Ensure all site system roles are correctly configured in the new environment.
 - Check for any missing or incorrect settings.
2. **Network Connectivity:**
 - Verify network settings and connectivity between the source and destination hierarchies.
 - Check firewall rules and DNS settings.

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3. **Data Integrity:**
 - Validate that all data has been migrated correctly.
 - Check for any data corruption or loss during the migration.
4. **Logs and Monitoring:**
 - Review SCCM logs for any errors or warnings.
 - Use monitoring tools to track system performance and identify issues.
5. **Client Communication:**
 - Ensure clients can communicate with the new management points.
 - Check client logs for any communication issues.
6. **Performance Issues:**
 - Identify and resolve any performance bottlenecks.
 - Optimize system resources and configurations for better performance.

Here are some specific SCCM log files that can help you troubleshoot various issues:

Client Installation Issues

1. **ccmsetup.log:** Records the client installation process.
2. **client.msi.log:** Details about the client MSI installation.

Software Distribution Issues

1. **execmgr.log:** Records details about software distribution and package execution.
2. **AppEnforce.log:** Records details about application enforcement.

Software Update Issues

1. **WUAHandler.log:** Interaction with the Windows Update Agent.
2. **UpdatesHandler.log:** Processing of software updates.
3. **UpdatesDeployment.log:** Details about software updates on the client.

Operating System Deployment Issues

1. **smsts.log:** Records details about task sequence execution.
2. **osdsetup.log:** Records details about operating system deployment setup.
3. **osdapplyos.log:** Records details about applying the operating system image.

Content Management Issues

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1. **ContentTransferManager.log**: Manages content transfer.
2. **DataTransferService.log**: Records details about data transfer operations.
3. **PkgXferMgr.log**: Manages the transfer of content between sites.

Management Point Issues

1. **mpcontrol.log**: Records the management point installation and health status.
2. **MP_Location.log**: Records management point location requests.
3. **MP_Ddr.log**: Records the conversion of XML DDR records from clients into the database.

General Troubleshooting

1. **smsexec.log**: Records the processing of all site server component threads.
2. **distmgr.log**: Manages content distribution.
3. **hman.log**: Records site configuration changes.
4. **sender.log**: Records the sending of site-to-site data.

Session 1: Introduction to SCCM

- Overview of SCCM: What is SCCM? Key features and benefits.
- SCCM Architecture: Components and their roles.
- Installation and Configuration: Basic setup and configuration.
- SCCM Services: Detailed explanation of core services like Endpoint Protection, Software Update Management, and Health Monitoring.

Session 2: SCCM Environment High-Level Design (HLD)

- Design Principles: Best practices for designing an SCCM environment.
- HLD Components: Servers, databases, and network considerations.

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- Example HLD: Walkthrough of a sample high-level design.

Session 3: SCCM Deployment and Configuration

- Deployment Strategies: Planning and executing SCCM deployments.
- Configuration Management: Setting up and managing configurations.
- Custom Configuration Items: Creating and deploying custom configurations.
- PowerShell Scripts
- Task Scheduler Scripts

Session 4: WSUS and Patching Solution - Part 1

- Introduction to WSUS: What is WSUS? Key features and benefits.
- WSUS Integration with SCCM: How WSUS works with SCCM.
- Setting Up WSUS: Installation and configuration of WSUS.

Session 5: WSUS and Patching Solution - Part 2

- Patch Management: Overview of patch management in SCCM.
- Deploying Updates: Creating and deploying update packages.
- Compliance Reporting: Monitoring and reporting on patch compliance.

Session 6: SCCM Reporting, Monitoring, and Maintenance

- Reporting Services: Using SCCM reporting features.
- Custom Reports: Creating and customizing reports.
- Monitoring SCCM: Tools and techniques for monitoring SCCM environment.
- Common Issues: Identifying and resolving common SCCM issues.
- Maintenance Tasks: Regular maintenance tasks to keep SCCM running smoothly.
- Troubleshooting Tools: Tools and techniques for troubleshooting SCCM.

Session 7: Special Scripts and Jobs

- PowerShell Integration: Using PowerShell with SCCM.
- Automation Scripts: Common scripts for automation and management.
- Custom Jobs: Creating and managing custom jobs.

Session 8: Advanced Configuration and Customization

- Custom Configuration: Advanced configuration settings and customization.
- Role-Based Administration: Setting up and managing roles and permissions.
- Integration with Other Tools: Integrating SCCM with other Microsoft and third-party tools.

Session 9: Power BI Dashboard Creation

- Introduction to Power BI: Overview of Power BI and its capabilities.

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- Creating Dashboards: Steps to create dashboards from reports.
- Pinning Visuals: How to pin visuals to dashboards.
- Advanced Dashboard Features: Using advanced features like live tiles and custom visuals.
- Optimizing Dashboards: Best practices for optimizing dashboard performance.
- SCCM Custom Dashboard:
 - Overview of the custom dashboard with all servers' information.
 - Detailed walkthrough of the patching information included in the dashboard.
 - How to use the custom dashboard to improve services.

Session 10: Troubleshooting and Maintenance

- Common Issues: Identifying and resolving common SCCM issues.
- Maintenance Tasks: Regular maintenance tasks to keep SCCM running smoothly.
- Troubleshooting Tools: Tools and techniques for troubleshooting SCCM.

Session 11: Software Deployment and Management

- Application Management: Deploying and managing applications.
- Software Updates: Managing software updates and patches.
- Compliance Settings: Ensuring compliance with organizational policies.

Certainly! Here are some deep troubleshooting questions for SCCM that you might find useful:

1. **How do you troubleshoot client installation issues in SCCM?**
 - Discuss the steps to verify prerequisites, check client logs (e.g., ccmsetup.log, ClientIDManagerStartup.log), ensure network connectivity, and review boundaries and boundary groups.
2. **What steps would you take to resolve software distribution failures?**
 - Explain how to check the execmgr.log and AppEnforce.log on the client, verify content distribution status on the Distribution Point, and ensure the application/package is correctly configured.
3. **How do you handle issues with Software Update Point (SUP) synchronization?**

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- Describe the process of checking Wsyncmgr.log, WCM.log, and WSUSCtrl.log, verifying WSUS configuration, and ensuring network connectivity between the SUP and WSUS server.
- 4. **What are the common causes of SCCM client health issues and how do you resolve them?**
 - Discuss common issues such as WMI corruption, missing prerequisites, and misconfigured boundaries. Explain how to use the ccmrepair command and the Client Health Evaluation tool.
- 5. **How do you troubleshoot Operating System Deployment (OSD) failures?**
 - Detail the steps to review smsts.log, check Task Sequence configurations, verify network access to the Distribution Point, and ensure drivers are correctly injected.
- 6. **What methods do you use to diagnose and fix issues with SCCM reporting?**
 - Explain how to check rscorp.log, verify SQL Server Reporting Services (SSRS) configuration, and ensure the Reporting Services Point is correctly installed and configured.
- 7. **How do you troubleshoot boundary and boundary group issues in SCCM?**
 - Describe how to verify boundary configurations, ensure boundary groups are correctly associated with Distribution Points, and check client location logs like LocationServices.log.
- 8. **What steps would you take to resolve issues with SCCM site system roles?**
 - Discuss how to check role-specific logs (e.g., MPSetup.log for Management Point, distmgr.log for Distribution Point), verify role configurations, and ensure network connectivity.

Absolutely! Here are some additional deep troubleshooting questions for SCCM:

1. **How do you troubleshoot issues with SCCM client policy retrieval?**
 - Discuss how to check the PolicyAgent.log and PolicyEvaluator.log on the client, ensure the client is assigned to the correct site, and verify that the Management Point is functioning correctly.
2. **What steps would you take to resolve Distribution Point (DP) content distribution failures?**
 - Explain how to review distmgr.log, pkgxfermgr.log, and smsdpmon.log, check the status of the DP, and ensure there is sufficient disk space and network connectivity.
3. **How do you handle issues with SCCM database replication?**

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- Describe the process of checking rcmlntrl.log, replmnr.log, and sender.log, verifying SQL Server configuration, and ensuring network connectivity between sites.
- 4. **What methods do you use to diagnose and fix issues with SCCM client communication?**
 - Discuss how to check ClientLocation.log, LocationServices.log, and ClientIDManagerStartup.log, ensure the client can reach the Management Point, and verify boundary configurations.
- 5. **How do you troubleshoot SCCM Task Sequence failures?**
 - Detail the steps to review smsts.log, check Task Sequence dependencies, verify network access to the Distribution Point, and ensure drivers and packages are correctly configured.
- 6. **What steps would you take to resolve issues with SCCM site server performance?**
 - Explain how to monitor server performance using tools like Performance Monitor, review smsdbmon.log and smsexec.log, and ensure SQL Server is optimized and properly configured.
- 7. **How do you handle issues with SCCM role-based administration (RBA)?**
 - Describe the process of checking SMSAdminUI.log, verifying security roles and scopes, and ensuring the correct permissions are assigned to users.
- 8. **What methods do you use to diagnose and fix issues with SCCM reporting services?**
 - Discuss how to check rscorp.log, verify SSRS configuration, ensure the Reporting Services Point is correctly installed, and troubleshoot report execution errors.
- 9. **How do you troubleshoot issues with SCCM software metering?**
 - Explain how to review mtrmnr.log, ensure the Software Metering Client Agent is enabled, and verify that the metering rules are correctly configured and applied.
- 10. **What steps would you take to resolve issues with SCCM compliance settings?**
 - Describe the process of checking DCMAgent.log, verifying configuration baselines and items, and ensuring the client can communicate with the Management Point.

Certainly! Here are some advanced SCCM troubleshooting questions tailored for someone with 10 years of experience:

1. **How do you troubleshoot and resolve SCCM database performance issues?**
 - Discuss how to monitor SQL Server performance using tools like SQL Server Profiler, review smsdbmon.log, optimize indexes, and ensure proper SQL Server configuration and maintenance plans.
2. **What steps would you take to diagnose and fix issues with SCCM site-to-site replication?**
 - Explain how to check rcmlntrl.log, replmnr.log, and sender.log, verify SQL Server replication settings, and ensure network connectivity and bandwidth between sites.

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3. **How do you handle complex application deployment failures in SCCM?**
 - Detail the process of reviewing AppEnforce.log, AppDiscovery.log, and CAS.log, ensuring application dependencies are met, and verifying content distribution and detection methods.
4. **What methods do you use to troubleshoot SCCM client policy processing issues?**
 - Describe how to check PolicyAgent.log, PolicyEvaluator.log, and ClientIDManagerStartup.log, ensure the client is correctly assigned to a site, and verify Management Point availability.
5. **How do you resolve issues with SCCM Management Point (MP) communication?**
 - Discuss how to review mpcontrol.log, mpmsi.log, and mp_hinv.log, ensure IIS configuration is correct, and verify network connectivity and firewall settings.
6. **What steps would you take to troubleshoot SCCM Distribution Point (DP) content validation failures?**
 - Explain how to check distmgr.log, pkgxfermgr.log, and smsdpmon.log, verify content library integrity, and ensure the DP has sufficient disk space and network connectivity.
7. **How do you handle issues with SCCM Endpoint Protection deployment and management?**
 - Describe the process of reviewing EPMgr.log, EndpointProtectionAgent.log, and AMPolicy.log, ensuring definition updates are correctly distributed, and verifying policy application.
8. **What methods do you use to diagnose and fix SCCM reporting services issues?**
 - Discuss how to check rscorp.log, verify SSRS configuration, ensure the Reporting Services Point is correctly installed, and troubleshoot report execution errors.
9. **How do you troubleshoot SCCM Task Sequence (TS) execution failures?**
 - Detail the steps to review smsts.log, check Task Sequence dependencies, verify network access to the Distribution Point, and ensure drivers and packages are correctly configured.
10. **What steps would you take to resolve issues with SCCM compliance settings and baselines?**
 - Describe the process of checking DCMAgent.log, verifying configuration baselines and items, and ensuring the client can communicate with the Management Point.

Client Installation Errors

Sure, here are some steps to troubleshoot SCCM (System Center Configuration Manager) client installation issues:

1. **Check Prerequisites:**
 - Ensure the target machine meets all prerequisites, such as having the correct version of Windows Installer and BITS (Background Intelligent Transfer Service) installed
2. **Review Log Files:**
 - **ccmsetup.log:** Located in C:\Windows\ccmsetup\Logs, this log file provides details about the client installation process.
 - **client.msi.log:** Located in the same directory, this log file contains information about the MSI installation process
3. **Verify Network Connectivity:**
 - Ensure the client machine can communicate with the SCCM server. Check for any network issues or firewall settings that might be blocking communication
4. **Check Client Push Installation Account:**
 - Ensure the account used for client push installation has administrative rights on the target machine
5. **Review Boundaries and Boundary Groups:**
 - Verify that the client machine is within the defined boundaries and boundary groups in SCCM
6. **Check Disk Space:**
 - Ensure there is sufficient disk space on the client machine for the installation
7. **Temporarily Disable Antivirus:**
 - Sometimes antivirus software can interfere with the installation process. Temporarily disable it and try the installation again
8. **Manually Install the Client:**
 - If the push installation fails, try manually installing the client using the ccmsetup.exe file
9. **Reset SCCM Client Policies:**
 - Reset the client policies to ensure they are correctly applied
10. **Check Group Policy Settings:**
 - Ensure that group policies are not blocking the installation or communication with the SCCM server

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Here is some common SCCM client installation error codes and their meanings:

1. **Error Code 2:** The system cannot find the file specified. This often occurs when the WMI service is corrupt
2. **Error Code 5:** Access denied. Ensure the installation account is a member of the Administrator Group
3. **Error Code 7:** Reboot required
4. **Error Code 8:** Setup already running
5. **Error Code 9:** Prerequisite evaluation failure
6. **Error Code 10:** Setup manifest hash validation failure
7. **Error Code 52:** Duplicate name exists on the network. Check for duplicate names in DNS
8. **Error Code 0x87D00202:** Service is shutting down
9. **Error Code 0x87D00207:** Parsing error
10. **Error Code 0x87D00213:** Timeout occurred
11. **Error Code 0x87D00215:** Item not found
12. **Error Code 0x87D00324:** Application was not detected after installation completed
13. **Error Code 0x87D00607:** Content not found

Here are some steps to troubleshoot SCCM content distribution issues:

1. **Check Distribution Point Configuration:**
 - Ensure the distribution point (DP) is correctly configured and has sufficient disk space
2. **Review Log Files:**
 - **DistMgr.log:** Located on the site server, this log file provides information about the distribution manager's activities.
 - **PkgXferMgr.log:** Also on the site server, this log file details the package transfer manager's activities.
 - **SMSDPProv.log:** Located on the DP, this log file shows the status of content being added to the content library
3. **Verify Package Properties:**
 - Ensure the package is correctly configured and associated with the correct DP
4. **Check Network Connectivity:**
 - Verify that the site server and DP can communicate without any network issues or firewall blocks
5. **Redistribute Content:**
 - If content distribution fails, try redistributing the content to the DP

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6. **Repair WMI on the DP:**
 - Sometimes, issues with the Windows Management Instrumentation (WMI) on the DP can cause distribution problems. Repairing WMI might help
7. **Check Permissions:**
 - Ensure that the necessary permissions are set correctly on the DP and the site server
8. **Restart ConfigMgr Services:**
 - Restarting the Configuration Manager services on the site server and DP can sometimes resolve distribution issues
9. **Review Content Library:**
 - Check the content library on the DP to ensure that the content is present and correctly referenced
10. **Database Replication:**
 - If multiple sites are involved, ensure that database replication is working correctly and that the database links between relevant sites are active

To check the permissions for a Distribution Point (DP) in SCCM, follow these steps:

1. **Open the SCCM Console:**
 - Launch the Configuration Manager console on your site server.
2. **Navigate to Administration:**
 - Go to the **Administration** workspace.
3. **Select Distribution Points:**
 - In the **Administration** workspace, expand **Site Configuration** and select **Servers and Site System Roles**.
4. **Choose the DP:**
 - Select the server that hosts the DP role. In the details pane, right-click the **Distribution Point** role and select **Properties**.
5. **Review Permissions:**
 - In the **Properties** window, go to the **Security** tab. Here, you can review and modify the permissions for the DP. Ensure that the necessary accounts have the appropriate permissions, such as **Read**, **Copy to Distribution Point**, and **Modify**
6. **Check NTFS Permissions:**
 - On the DP server, check the NTFS permissions on the content library folder (typically located at C:\SCCMContentLib). Ensure that the **SYSTEM** account and the **SCCM server** have full control
7. **Verify IIS Permissions:**
 - If IIS is used, ensure that the **IIS_IUSRS** group has the necessary permissions on the DP's virtual directories

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Here are some common issues you might encounter with SCCM content distribution:

1. **Network Connectivity:**
 - Ensure that the site server and distribution point (DP) can communicate without any network issues or firewall blocks
2. **Insufficient Disk Space:**
 - Verify that the DP has enough disk space to store the content
3. **Permissions Issues:**
 - Ensure that the necessary permissions are set correctly on the DP and the site server
4. **WMI Issues:**
 - Problems with the Windows Management Instrumentation (WMI) on the DP can cause distribution issues. Repairing WMI might help
5. **Content Library Corruption:**
 - Check the content library on the DP to ensure that the content is present and correctly referenced
6. **Database Replication:**
 - If multiple sites are involved, ensure that database replication is working correctly and that the database links between relevant sites are active
7. **Log File Errors:**
 - Review log files such as **DistMgr.log**, **PkgXferMgr.log**, and **SMSDPProv.log** for any errors or issues
8. **Configuration Issues:**
 - Ensure the DP is correctly configured and associated with the correct package

Content library corruption in SCCM (System Center Configuration Manager) refers to issues where the content library on a distribution point (DP) becomes damaged or contains invalid data. The content library is a critical component that stores all the files needed for software deployments, such as applications, software updates, and operating system images

Causes of Content Library Corruption

1. **Disk Errors:** Physical disk errors or bad sectors can corrupt the content library files
2. **Network Issues:** Interrupted or unstable network connections during content transfer can lead to incomplete or corrupted files
3. **Permissions Issues:** Incorrect permissions on the content library folders can prevent proper access and updates, leading to corruption
4. **WMI Issues:** Problems with the Windows Management Instrumentation (WMI) on the DP can cause issues with content management

Symptoms of Content Library Corruption

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- **Failed Content Distribution:** Content fails to distribute to the DP, and errors are logged in files like **DistMgr.log** and **PkgXferMgr.log**
- **Missing or Invalid Data:** Content is missing or invalid, causing deployment failures
- **Validation Errors:** Running validation checks on the content library reveals errors or inconsistencies

Troubleshooting Steps

1. **Run Validation Checks:**
 - Use the SCCM console to validate the content library. This process checks for missing or corrupt files and helps identify issues
2. **Review Log Files:**
 - Check log files such as **DistMgr.log**, **PkgXferMgr.log**, and **SMSDPProv.log** for any errors or warnings related to content distribution
3. **Use Content Library Explorer Tool:**
 - The Content Library Explorer tool allows you to explore and validate the content library on a specific DP. It can help identify and fix issues with the content
4. **Redistribute Content:**
 - If corruption is detected, redistribute the affected content to the DP. This process ensures that the content is correctly copied and validated
5. **Repair WMI:**
 - If WMI issues are suspected, repair the WMI repository on the DP to ensure proper functioning
6. **Check Disk Health:**
 - Run disk checks to identify and fix any physical disk errors that might be causing corruption

WSUS Search Fix

```
# Stop windows update service
Stop-Service -Name wuauserv -Force
Write-Output "Windows Update service stopped."

# Stop Background Intelligent Transfer Service (BITS)
Stop-Service -Name bits -Force
Write-Output "BITS service stopped."

# Delete contents of SoftwareDistribution folder
$SoftwareDistributionPath = "C:\Windows\SoftwareDistribution"
Remove-Item -Path "$SoftwareDistributionPath\*" -Recurse -Force
Write-Output "Contents of SoftwareDistribution folder deleted."

# Start windows update service
Start-Service -Name wuauserv
Write-Output "Windows Update service started."

# Start Background Intelligent Transfer Service (BITS)
Start-Service -Name bits
Write-Output "BITS service started."
```